



OPERATION Department
SIMHADRI



Ref: SMTTP/OPERATION/2023-24/opn/139658

Date:

19/05/2023(18:48:39)

Subject: LMI on GUIDELINES FOR UNIT UNDER RESERVE
SHUTDOWN

LMI on "GUIDELINES FOR UNIT UNDER RESERVE SHUTDOWN" LMI/OGN/OPS/SYST/038

The following LMI has been revised in line with revised OGN vide "COS-ISO-00-OGN/OPS/SYST/038 Rev. No: 01, April 2023"

Revision details:

- Guidelines for CHP and AHP Added for units under RSD.

Put up for kind approval please

Submitted by:

जुलियट जॉनी - इंजीनियर

JULIET JOHNY - 105435 (ENGINEER ,OPERATION) ,Simhadri
(JULIETJOHNY@NTPC.CO.IN, 7597808553)

On: 19/05/2023(18:48:39) **Ref:** SMTTP/OPERATION/2023-24/opn/139658

May Please review & approve

Submitted by:

एन दुर्गा नायक - वरिष्ठ प्रबंधक

NENAVATHU DURGA NAIK - 100440 (SENIOR MANAGER ,OPERATION)
(NENAVATHUDURGANAIAK@NTPC.CO.IN, 8331019130) ,Simhadri

On:20/05/2023(08:38:38) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

May please review.

Submitted by:

टी वी राजशेखर - अपर महाप्रबंधक

T V Rajasekhar - 007005 (ADDL. GENERAL MANAGER ,OPERATION)
(TVRSEKHAR@NTPC.CO.IN, 9650998176) ,Simhadri

On:20/05/2023(18:40:17) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:23/05/2023(17:17:14) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

May pl. be approved

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MECH MAINT- BOILER)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:24/05/2023(09:25:17) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Review

Submitted by:

शंभू कुमार सुमन - अपर महाप्रबंधक

Shambhu Kumar Suman - 005007 (ADDL. GENERAL MANAGER ,MECH MAINT- TURBINE)
(SKSUMAN@NTPC.CO.IN, 9431215296) ,Simhadri

On:24/05/2023(18:17:03) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

LMI reviewed.

Forwarded for further approval.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

RASHMI RANJAN SATAPATHY - 100837 (SENIOR MANAGER ,MECH MAINT- TURBINE)
(RRSATPATHY@NTPC.CO.IN, 9437021584) ,Simhadri

On:31/05/2023(11:11:33) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Submitted by:

धर्मेन्द्र कुमार तोमर - अपर महाप्रबंधक

Dharmendra Kumar Tomar - 007286 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(DHARMENDRAKUMAR02@NTPC.CO.IN, 9416212079) ,Simhadri

On:01/06/2023(20:00:16) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

EMD side reviewed.

Put up for approval

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:07/06/2023(10:45:22) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

Pl review

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:07/06/2023(11:49:39) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Reviewed and put up for further approval.

Submitted by:

जी एच राव - अपर महाप्रबंधक

G H Rao - 006589 (ADDL. GENERAL MANAGER ,C & I MAINT- BOILER)
(GHNRAO@NTPC.CO.IN, 8331016245) ,Simhadri

On:07/06/2023(18:30:22) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

TR Rajagopalan - 003994 (ADDL. GENERAL MANAGER ,C & I MAINT)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:08/06/2023(12:27:02) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

Please review

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:08/06/2023(18:03:19) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

Please refer to Annexure- LMI revised file "O&M Guidelines of Unit RSD revision 2 may 23. pdf" , page no. - 26

ASH HANDLING SYSTEM.

For "Condition of Refractory and mother plate" checking, to execute the work S-Panel sealing is needed along with scaffolding erection in BA hopper by BMD.

Please note that it is a time consuming process.

Submitted by:

तन्मय सिन्हा - वरिष्ठ प्रबंधक

TANMOY SINHA - 101612 (SENIOR MANAGER ,ASH HANDLING)

(TANMOYSINHA@NTPC.CO.IN, 9434737629) ,Simhadri

On:09/06/2023(11:18:28) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Refractory inspection shall be carried out after S-panel sealing by BMD & subsequent scaffold erection as time permits.

May be approved

Submitted by:

नीलांजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)

(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:09/06/2023(15:58:33) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

May pl review

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:10/06/2023(18:19:01) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

LMI reviewed. Forwarded for further approval.

Submitted by:

सुनीता गुलीपिल्ली - प्रबंधक

SUNEETA GULIPILLI - 101282 (MANAGER ,CHEMISTRY)
(SUNEETAGULIPILLI@NTPC.CO.IN, 9493547082) ,Simhadri

On:22/06/2023(16:12:09) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Reviewed and forwarded for further approval.

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:22/06/2023(21:40:35) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Submitted by:

राजेश चावला - अपर महाप्रबंधक

Rajesh Chawla - 007383 (ADDL. GENERAL MANAGER ,MTP)
(RAJESHCHAWLA@NTPC.CO.IN, 9425534528) ,Simhadri

On:23/06/2023(17:25:18) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Submitted by:

विश्व मोहन चौधरी - महाप्रबंधक

Vishwa Mohan Choudhary - 003932 (GENERAL MANAGER ,FUEL MANAGEMENT)
(VMCHOUDHURY@NTPC.CO.IN, 7588691232) ,Simhadri

On:23/06/2023(17:53:13) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

May be accorded approval please.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

Mathew Eipe Kovoor - 005637 (GENERAL MANAGER ,MAINTENANCE)
(MEKOVOOR@NTPC.CO.IN, 9650994652) ,Simhadri

On:03/07/2023(19:04:04) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Please review Page no-7 (Table)

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:04/07/2023(11:19:37) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Reviewed. Forwarded for approval.

Submitted by:

के कृष्णा रेड्डी - अपर महाप्रबंधक

K KRISHNA REDDY - 032588 (ADDL. GENERAL MANAGER ,OPERATION)
(KKRISHNAREDDY@NTPC.CO.IN, 9440901769) ,Simhadri

On:14/07/2023(09:17:43) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

1. Pl incorporate/revise the LMI as per the AGM(AHP) comments in the pre page.
2. The LMI may be routed to AGM(CHP) I/c also for noting the revision w.r.to CHP area

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

TR Rajagopalan - 003994 (GENERAL MANAGER ,OPERATION)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:14/07/2023(11:17:00) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Necessary changes have been made as proposed by AHM.

Details of changes: Page no 24 under ASH HANDLING SYSTEM

For " Condition of refractory and mother plate, S-panel sealing is needed along with scaffolding erection by BMD"

Please review.

Submitted by:

एन दुर्गा नायक - वरिष्ठ प्रबंधक

NENAVATHU DURGA NAIK - 100440 (SENIOR MANAGER ,OPERATION)

(NENAVATHUDURGANAIAK@NTPC.CO.IN, 8331019130) ,Simhadri

On:15/07/2023(11:31:10) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

LMI reviewed & forwarded for approval.

Submitted by:

तन्मय सिन्हा - वरिष्ठ प्रबंधक

TANMOY SINHA - 101612 (SENIOR MANAGER ,ASH HANDLING)

(TANMOYSINHA@NTPC.CO.IN, 9434737629) ,Simhadri

On:15/07/2023(11:46:23) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

Submitted by:

नीलाजन मंडल - अपर महाप्रबंधक

Nilanjan Mandal - 007418 (ADDL. GENERAL MANAGER ,ASH HANDLING)
(NILANJANMANDAL@NTPC.CO.IN, 9434097418) ,Simhadri

On:17/07/2023(17:57:24) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

LMI reviewed.

Submitted by:

सन्जित मजूमदार - अपर महाप्रबंधक

Sanjit Majumdar - 005065 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(SMAZUMDAR@NTPC.CO.IN, 9650990065) ,Simhadri

On:17/07/2023(18:28:42) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

May please review & approve LMI

Submitted by:

एन दुर्गा नायक - वरिष्ठ प्रबंधक

NENAVATHU DURGA NAIK - 100440 (SENIOR MANAGER ,OPERATION)
(NENAVATHUDURGANAIAK@NTPC.CO.IN, 8331019130) ,Simhadri

On:18/07/2023(09:04:44) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

As recorded by GM(O&M) in Pg 11 of the proposal, "please review Page no-7 (Table)" and put up for further approval

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

TR Rajagopalan - 003994 (GENERAL MANAGER ,OPERATION)
(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:18/07/2023(09:58:20) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

LMI reviewed by all sectional HOD s' , may please be approved.

Submitted by:

एन दुर्गा नायक - वरिष्ठ प्रबंधक

NENAVATHU DURGA NAIK - 100440 (SENIOR MANAGER ,OPERATION)

(NENAVATHUDURGANAIAK@NTPC.CO.IN, 8331019130) ,Simhadri

On:19/07/2023(14:20:45) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

The revised LMI has been reviewed by all concerned sectional Heads and wherever comments are there, changes have been incorporated.

The LMI is forwarded for approval

Submitted by:

टी.आर. राजगोपालन् - महाप्रबंधक

TR Rajagopalan - 003994 (GENERAL MANAGER ,OPERATION)

(TRRAJAGOPAL@NTPC.CO.IN, 9440918024) ,Simhadri

On:19/07/2023(17:48:39) **Ref:**SMTPP/OPERATION/2023-24/opn/139658

May please approve.

Submitted by:

धुरजति प्रसाद पात्रा - महाप्रबंधक

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:20/07/2023(10:28:41) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:25/07/2023(13:22:58) **Ref:**SMTTP/OPERATION/2023-24/opn/139658

Local Management Instructions

GUIDELINES FOR UNIT UNDER RESERVE SHUTDOWN



Approved for implementation by HEAD OF PROJECT

Date 18.05.2023

Prepared by OPERATION DEPARTMENT

NTPC LTD.

Simhadri Super Thermal Power Project.

GUIDELINES FOR UNIT UNDER RESERVE SHUTDOWN

LMI	<i>Guidelines for unit under Reserve shutdown</i>		
LMI No	LMI/OGN/OPS/SYST/038		
Issue No	1	Revision: 2	Dt: 18.05.2023

	Name	Designation	Signature
Prepared by:	JULIET JOHNY	Engineer (O)	
Checked by:	N Durga Naik	Senior Manager (O)	
Checked by:	T V RAJASEKHAR	AGM (O)	
Checked by	Ajay Gupta	DGM (EEMG)	
Checked by	OM PRAKASH	AGM (BMD)	
Checked by	S.K. SUMAN	AGM (TMD)	
Checked by	D.K.TOMAR	AGM(EMD)	
Checked by	T.R. RAJAGOPALAN	AGM (C&I)	
Checked by	NILANJAN MANDAL	AGM (MM-AHM)	
Checked by	PIYUSH SRIVASTAVA	AGM(CHEM)	
Checked by	SANJIT MAJUMDAR	AGM(CHP)	
Checked by	RAJESH CHAWLA	AGM (MTP)	
Rechecked by	MATHEW EIPE KOVOOR	GM (Maint)	
Reviewed by:	D. P. PATRA	GM (O&M)	
Approved by	SANJAY KUMAR SINHA	HOP (Simhadri)	

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Guidelines for unit under Reserve shutdown

1.0	INTRODUCTION
	Now a day's thermal power plant have been challenged with protecting equipment during shutdown periods for months and years. However, recent industry drivers have created new scenarios and circumstances, increasing the importance of laying up plants for extended periods of time.
	Shutdown durations are driven by many factors and are seen when a unit is in "no demand" or "economic outage." Generally, the exact day of return to service is not known. However, based on system operations and plant specific requirements, units are required to return to service in a timely manner when called on. These shutdowns are usually not a planned outage in which equipment will be opened for maintenance. Instead, systems remain intact while the plant is waiting to be called on.
	Many times these situations are complicated in the early stages of the shutdown, especially the first couple of weeks, as different steps in the layup process may be delayed by uncertainties of when the unit will be called on. This leads to individual systems, often support systems, being left in an uncontrolled status, just shut down, or left in service for extended periods without consideration for the best layup technique
	This LMI aims towards the uniform code of practice to be followed while units are under reserve shut down and forced outage during which the healthiness of equipment needs to be maintained. The aim of this LMI is to ensure equipment is available whenever they are needed to start –up. Thus, the procedures have been made for Lay-up with intent to keep all systems and related equipment in healthy condition ready to start and operate at any time when needed.
2.0	SUPERSEDED DOCUMENTS & REVISION NOTES
	Superseded document : LMI/OGN/OPS/SYST/038 rev:1 Nov 22 Revision Notes: Guidelines for CHP and AHP Added for units under RSD.
3.0	SCOPE:
	This LMI document is applicable for 4 X 500 MW Units of Simhadri STPP, Stage-I & II units.
4.0	GOALS OF LAY-UP :
	The principal goal of proper power plant layup is to maintain the plant in a state of preservation that protects important systems and equipment and allows return to service without compromising plant integrity. This document is built on fundamental goals of fossil and gas plant layup as follows: 1. Ensuring Reliability. 2. Optimization of resources. 3. Safety remains paramount. 4. Environmental Concerns addressed

5.0	LAYUP TECHNIQUES
	To achieve the goals of layup described previously, various layup techniques may be used that best align with an individual station's Applicability. Although common terminology may imply that a unit is layed up "wet" or "dry," it is essential to evaluate each system on a case-by-case basis and determine the proper state of layup for each system/component. This may include a combination of the following techniques: • Wet layup • Wet layup with a nitrogen blanket • Dry layup with a dehumidified air or nitrogen blanket • Drained • Shutdown as is • Keep in service • Maintain circulation • Surface barriers • Stored
5.1	WET LAYUP
	Wet layup is a primary layup technique for boilers and feed water and condensate systems where chemistry will be maintained in such a condition to prevent corrosion. The boiler will be full of water. If treated with high doses of ammonia (& hydrazine for AVT-R units), it may be necessary to drain and refill a boiler in cold standby before filling for start-up. Wet layup in the feed water and condensate system equipment generally consists of filling the components and connecting piping with demineralized water that contains the proper chemicals for the metallurgy of the system. The ability to circulate, monitor, and add chemicals as required during shutdown periods becomes increasingly important for preservation of the equipment as the duration of the layup period increases.
5.2	WET LAYUP WITH A NITROGEN BLANKET
	Simhadri shall use the wet layup method with the addition of a nitrogen gas blanket above the liquid level in the component or piping. The main advantage of wet layup with a nitrogen blanket is that it provides a more positive method for protecting equipment than some other methods do. The main disadvantage of this method is that requires more work by the operations and maintenance personnel to ensure that the nitrogen cap is not lost and that the chemical concentrations remain at the proper level. Note there are many different nitrogen application methods that can be applied depending on the unit design and there are significant advantages to determining methods that maximize deoxygenation (maximize nitrogen/water contact during introduction.
5.3	DRY LAYUP WITH AN AIR OR NITROGEN BLANKET
	Dry layup is an alternative for protecting closed-in equipment, such as pressure vessels and piping. Two options can be used for dry layup: one uses nitrogen gas to blanket the internal parts of the equipment and the other uses dry air. Nitrogen blanketing is generally not used for long-term layup because of the maintenance and operational requirements to keep the gas in the systems. The nitrogen blanketing method is more effective when protecting an airtight vessel for short periods. This layup method involves draining the system or component with a

	nitrogen cover gas, which prevents the ingress of air into the system or component during draining and maintains an inert gas environment during the layup period. Dry or dehumidified air offers the same advantages as nitrogen gas plus others. The dry condition can be maintained even while the equipment is opened for routine inspections. This allows routine inspections and repairs to take place without the loss of costly gas or possible corrosion taking place during the inspection or repair. Dehumidified air systems are also less costly to maintain over the long term. As Simhadri is situated at coastal area, nitrogen gas should be used for long term preservation of boiler and auxiliaries, as far practically feasible.
5.4	DRAINED
	Some systems are drained for the shutdown without maintaining any control of the layup environment, which exposes them to air throughout most of the shutdown period. This practice will not typically cause serious harm if the system materials are corrosion resistant unless deposits or crevices are present. When draining a system for layup, care should be taken to ensure that all of the system low points are cleared and no water, sludge, or deposits are left in piping that can create a corrosion-rich environment. This may require disassembly at certain low points. Systems that are drained should be cleaned and purged as needed to prevent piping degradation from taking place. Simhadri can use draining as a layup method for cooling water systems, ash-handling systems, and environmental controls systems. Draining may also be deemed appropriate should freezing be a concern during the layup period.
5.5	SHUTDOWN AS IS
	In the shutdown as-is layup condition, the system or component remains in the shutdown condition with no special treatment. If the system or component is filled with water during plant operation, it remains filled with water with no additional chemical treatment. The water environment will generally degrade due to the ingress of air. If a system contains steam during power operation, the steam is replaced by air and residual condensate, which is essentially equivalent to the drained layup method. This method can be used on a majority of plant systems for a period of time. However, the risk of this technique increases with time due to ingress of air that leads to oxidation within the system.
5.6	KEEP IN SERVICE
	Keeping selected equipment and systems in operation instead of shutting them down is another layup technique that may be used. Systems such as the turbine lubricating oil and fuel oil transfer systems contain fluids that provide a natural protection. The closed cooling water system can be left in operation because the cooling water has been treated with a corrosion inhibitor to give it protection. Digital control systems and control systems are also commonly left in service during the layup period. The main disadvantages with this technique are energy consumption, resources to monitor equipment operation, and the need for minor maintenance to keep them operating. However, these costs can be minor when compared with the alternatives to protect the same equipment for an extended period.
5.7	MAINTAIN CIRCULATION
	Many plants have found that the best form of layup for cooling water and air systems is to maintain some type of circulation. Several forms of localized corrosion are more likely to occur when a system is left stagnant or is drained and unprotected. This technique is different from keeping systems in service because it typically requires an abnormal lineup or addition of temporary modification to allow for system circulation.

5.8	SURFACE BARRIERS									
	Primers, paints, or preservative coatings can be used to protect most surfaces. Proper barrier surfaces can offer protection from corrosion, rot, and deterioration from causes such as corrosive vapors, sunlight, marine growth, and galvanic action. These methods can be used during the layup period to protect external surfaces such as structures and other equipment that can be exposed to the elements of the weather. The main disadvantage of this method is that it requires extensive effort to prepare the surfaces and apply the proper barrier surfaces. Internal surface barriers can also be an effective means of protecting plant equipment. There are treatments that provide equipment protection by establishing a barrier between the oxide surface and any water or moisture that may be present.									
5.9	STORED									
	Some plant components and systems should be placed in a stored condition for longer duration shutdowns. This tends to be applicable to many electrical components. Switchgear may be dried out and wrapped with poly plastic sheets to protect it from the environment. Desiccant may be placed in cubicles and changed out regularly to absorb moisture. Large breakers may be racked out and stored in place. Motors may be removed and placed in clean, dry locations in a building.									
6.0	DETERMINING THE CORRECT LAYUP TECHNIQUE									
System	Sub system	Wet lay up	Wet with N2	Dry with air	Dry with N2	Drained	As is	Keep in service	Maintain circulation	Barrier surfaces
Boiler systems	Boiler fire side			✓	✓					
	Boiler Tube internals	✓	✓							
	Gas path			✓			✓			
	APHs			✓			✓			
	ID fans			✓			✓			
Fuel system & air	Mills			✓			✓			
	PA fans			✓			✓			
	FD fans			✓			✓			
	AHS-dry			✓			✓			
	AHS-wet					✓	✓			
Steam turbine & Generator	Turbine			✓			✓			
	Generator			✓	✓		✓			
	Exciter			✓	✓		✓			
	PW system						✓	✓		
	TG Lub oil						✓	✓		
	Seal oil						✓	✓		
	H2 Coolers						✓	✓		
Feed water & condensate	Condenser & Hot well	✓	✓	✓		✓	✓			
	FW & condensate	✓	✓	✓		✓	✓			
	BFPs					✓	✓	✓		
	Heaters	✓	✓	✓		✓	✓			

	D/A	✓	✓			✓	✓			
BoP	CW					✓	✓	✓	✓	
	CT					✓	✓	✓	✓	
	Air sys						✓	✓		
	Fire water							✓		
	Service water							✓		
	Actuators						✓	✓		
Electric al system	SWGR						✓	✓		
	Transform ers						✓	✓		
	Motors (H)						✓	✓		
	Motors without heaters			✓			✓	✓		
	UPS						✓	✓		
	DCS						✓	✓		
C&I	Control system						✓	✓		
	SWAS						✓	✓		

7.0	PROCEDURE & LAY-UP MATRIX
	Activities that are infrequently performed, such as layup, increase the risk of human performance errors and can be performed more reliably with procedures and instructions. most importantly, the station should not expect to enter an entirely new set of operating conditions, albeit most systems are shutdown, without appropriate guidance for the station staff in the form of written procedures and instructions.
7.1	PRE-SHUTDOWN ACTIVITIES
	If unit is to be stopped is for RSD, following activities are to be planned. - Get the information of tentative time of RSD. - Inform CHP not to feed coal in coal bunkers. - MTP to be informed to plan for long pending jobs that needed unit shutdown. - As far as possible empty the coal feeders. - Plan for ATT of turbine. - Plan for actual turbine over speed, if it is due. - NO parameter reduction is required before stopping of Unit. - Ensure adequate level in fire water tank. - Close CRH to PRDS valve & ensure seal steam temp is adequate(>280 deg c).Raise PRDS pressure to 12 ksc - Normalize MS-1&2 (boiler stop valve) power supply. - Open CBD to flush out contaminants concentration in drum.
	RESPONSIBILITY (b to k):OPN/EEMG/MTP RESPONSIBILITY (a): EEMG in coordination with ROS/CC-OS/SIIS
7.2	SHUTDOWN ACTIVITY

	1. Empty the feeders from the top mills. 2. Revise the ON bar & OFF bar DC & accordingly reduce the load. 3. Take MDBFP in service & unload one TDBFP. 4. Carry out ATT half an hour prior to unit shutdown. 5. Ensure 3 continue mills (preferably B, C & D) in service. 6. Reduce coal flow such that feed rate in 3 mills is 40 tph 7. Charge HP/LP bypass & reduce drum pressure before stopping the unit to 70-75 ksc. 8. Bring unit load to 100 MW. 9. Stop the unit. 10. Record turbine coasting down time. 11. Post stopping the unit, Stop MDBFP at the earliest. 12. Purge the Boiler & stop one set of fans.
	RESPONSIBILITY:OPERATION/EEMG/ C&I
7.3	POST STOPPING ACTIVITIES
	1. Close boiler stop valves & depressurize MS line. 2. Close HPBP spray manual vale. 3. Close SH/RH spray manual valve. 4. Equipment to be stopped as per the stopping sequence after unit stoppage. 5. Close oil elevation manual valve of HONV & main isolation valve of HFO & LDO. 6. Carry out LRSB & wall blowing to depressurize the Boiler at the earliest. Once Flue gas temp at APH inlet comes to 200 deg c., take air motors in service & stop all fans to carryout APH water washing.
	RESPONSIBILITY:UCE
7.4	LAY-UP ACTIVITY
7.4.1	BOILER
	Boiler should <i>not be</i> cooled by feeding & draining & allowed to be cooled naturally after FG inlet temp at APH comes below 200 deg c. As coastal plants boilers are to be preserved in <i>wet mode only</i> , Boiler to be kept pressurized as long as possible. No vents or drains need to be opened.
	RESPONSIBILITY:UCE
	Once the boiler gets depressurized drain the boiler & water to be filled in boiler through SH only (backfilling of drum from SH circuit) with elevated pH (>10).
	RESPONSIBILITY--CHEMISTRY/OPERATION
	Provision of connecting N2 cylinders in preservation circuit to be made through.
	RESPONSIBILITY (BMD-PP)
	Back Filling through boiler fill pump to be done only after SH header metal temp comes below 100 deg c. (<i>approx. after 4 days</i>). Stop filling at drum pressure 10 ksc & connect N2 cylinder for capping. If N2 pressure is not holding, keep boiler pressurized up to 10 ksc. Keep BCW pumps in service for 30 min once in a week. pH in the boiler to be maintained 10-10.5. RH circuit to be preserved with N ₂ .
	RESPONSIBILITY—OPERATION/ CHEMISTRY/BMD
	Water sample to taken once in three days for Ph. value, If Ph. is less ammonia dosing to be done to improve Ph. (<i>ammonia can be dosed through PO4 dosing tank</i>).
	RESPONSIBILITY-CHEMISTRY
7.4.2	AIR PREHEATER

	APH water washing to be done till Ph. in sample come to 7.0.Keep APHs in service till they get completely dried (24 hr. after completion of water washing).If water washing is not planned, APHs to be stopped only after FG inlet temp comes below 60 deg C.Take out oil carry over probe & isolate cooling water. All APHs trial to be taken from main motor & air motor once in a day. APHs lamps healthiness to be ensured. Take trial run of bearing LOPs for 30 min once in a week.
	RESPONSIBILITY:UCE/BMD/AHD
	Once in week Gearbox vibration of all APHs to be measured.
	RESPONSIBILITY:MTP
	Perform visual inspection of baskets, seals & structures. Internal inspection to be done for any ash accumulation.
	RESPONSIBILITY:BMD-RF
	Take oil sample of GB & SB once in two weeks
	RESPONSIBILITY:CHEMISTRY
7.4.3	ID FANS
	ID fan internal inspection to be carried out. Ash accumulation, if any is there to be removed. Oil pump to be kept in service.ID fan trial to be taken once in 15 days. Oil sample to be tested preferably after fan trial run. ID fans inlet/ outlet gate operation to be checked & remote operation to be made through. If RSD gets extended, flue gas ducts inspection to be carried out.
	RESPONSIBILITY:OPN/BMD-RF
7.4.4	FD FAN
	Blade pitch operation of fans to be checked daily once. Fans to be run once in 15 days.Lub oil samples to be checked preferably after fans running. Fans internal inspection to be done. Vibration to be taken & recorded by MTP. Scanner air fan trial to be taken .If RSD gets extended secondary air duct inspection to be carried out.
	RESPONSIBILITY:OPN/BMD-RF/MTP
7.4.5	PA FANS
	Blade pitch operation of fans to be checked daily once. Fans to be run once in 15 days. Lube oil samples to be checked preferably after fans running. Fans internal inspection to be done. Seal air fans trial run also to be taken. Vibration to be taken & recorded by MTP. If RSD gets extended, Primary air ducts inspection to be carried out.
	RESPONSIBILITY:OPN/BMD-RF/MTP
7.4.6	MILLS
	Empty all the mills after it cools down. Take LOP trial once in a week. Take trial run of mills once in 15 days. Do PM Job of Mills. Oil sample to be taken once in 15 days.
	RESPONSIBILITY:OPN/BMD-RF/MTP/CHEMISTRY
7.4.7	RC FEEDERS
	Close all Feeder inlet & outlet gates. Empty the feeder .Feeder no load trial should preferably be taken once in 15 days. Bunker temp to be taken once in a week & to be recorded.
	RESPONSIBILITY:OPN
7.4.8	HFO/LDO PUMPS
	LDO pumps trial to taken once in a week. HFO pump to be kept running as there is a likelihood of HFO line chocking if it is kept off for prolong time. Moreover HFO takes more time to improve temp at burner front.
	RESPONSIBILITY:OPN
7.4.9	TURBINE

	1. TG to be allowed to cool naturally. 2. Equipment stopping sequence to be followed as per annex. 3. Once the shaft temp comes to 100 deg c, Turbine preservation to be started by giving dry air in HPT through fast cooling ports & RH to be maintained < 30%. 4. Hot well to be drained to prevent any moisture in TG. MOT centrifuge to be kept in service for 12 hr. in a day.
	RESPONSIBILITY:OPN/TMD/CHEMISTRY
7.4.9	CONDENSER VACUUM PUMPS
	Condenser vacuum pumps to be drained if RSD is extended. Trial to be taken once in 15 days.
	RESPONSIBILITY:UCE
7.4.10	CW PUMPS
	1. Post stopping of unit no CW pump should be stopped till HPC /HPT temp comes below 325 deg C. 2. Once HP shaft temp goes below 325 deg C., CW pumps can be stopped & condenser outlet valve can be kept closed to maintain static load of water box. 3. CW blow-down valve from this unit can be kept open so that circulating water flow is available for PHE cooling through ARCW. 4. If all CW pumps are in stopped condition, then fire water can be used for PHE cooling. Once in 3 days CW pumps to be kept in service for 2 hrs. For circulation of water in water box to avoid any vegetation growth OR Sea water make up and CW blow down interconnection can be kept opened for 3 hrs ,so that condenser flow can be established .
	RESPONSIBILITY:OPN
7.4.11	TDBFP
	1. TDBFPs barring can be stopped after casing temp comes to ambient temp. 2. Once in 7 days TDBFP to be put on barring for 6 hrs. If D/A is not drained else hand barring to be done once in 15 days. 3. TDBFP LOP trial to be taken once in 7 days. 4. TDBFP centrifuge to be kept in service daily for 12 hrs. 5. Oil sample to be taken once in 15 days.
	RESPONSIBILITY:UCE/CHEMISTRY
7.4.12	FEED WATER & CONDENSATE SYSTEM.
	1. D/a to be kept pressurized using Aux PRDS & feed water temp to be kept at 105-110 deg C 2. Maintain Feed water system & condensate system in wet lay-up & water chemistry to be maintained & sample to be taken once in a week.
	RESPONSIBILITY:UCE/CHEMISTRY
7.4.13	GENERATOR
	1. Seal oil system to be continued to keep in service & H2 pressure can be maintained 3.2-.3.25 ksc. & purity to be ensured. DC SOP trial to be taken as per standard practice. 2. Seal oil temp to be maintained within limits. (<45 deg C) 3. DMCW pumps to be kept running for seal oil cooling. 4. Generator PW system will be in service. Generator windings should not be kept bypassed from PW side for more than 48 hrs. 5. Ensure Generator shutdown heaters to be in auto.
	RESPONSIBILITY:UCE

7.4.14	MOTORS
	1. Ensure space heaters are in service during stopped condition. 2. Meggering of motors to be done before trial run. 3. Manual rotation of big motors to be considered if shutdown is extended. 4. Trial of Motors without space heaters to be taken once in 15 days.
	RESPONSIBILITY:OPN/EMD

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
BOILER SYSTEMS	BOILER FIRE SIDE	- Wall slagging is to be removed, particularly around burners. - Wall blower, LRSB may be operated after shutdown with available steam provided sufficient degree of superheat is available. - Ensure that the boiler pendant sections are cleared of slagging deposits. - Ash removal to prevent corrosion	- Inspect the boiler furnace area, bottom ash, pendent section (superheater, reheater, etc.) as soon as possible after unit is shutdown - Ensure that the soot blowers are not leaking through to the boiler including pendent area soot blowers.	
BOILER SYSTEMS	BOILER TUBE SIDE	- Wet or dry layup to be done for all boiler tubes as applicable. - If the unit will remain offline for an extended period, dry layup is the preferred method. Use boiler residual heat to ensure moisture is removed from all sections of boiler tubes - If the unit will remain in wet layup for days, use a nitrogen blanket on the boiler or ensure positive water pressure. - If the boiler is to be drained (other than BTL) then boiler should be drained as per OGN/OPS/CHEM/017 to ensure drying of water wall by latent heat and evaporation of any moisture in pendant coils. - Boiler fuel oil guns may be used as a heat source if the decision to move from wet to dry is made several days after shutdown of the unit.	- If dry layup is used, check relative humidity to ensure that dry conditions are maintained. - If a nitrogen blanket is used during layup, ensure that a positive nitrogen pressure is maintained on the boiler and check for oxygen concentration at each drain point. - During wet layup, ensure that appropriate water chemistry parameters are maintained in the boiler. Please refer the LMI OGN/OPS/CHEM/017 "Cycle chemistry guidelines for start-up, Shutdown and layup practices" for more details.	- Relative humidity should be measured in SH and RH circuit as per the frequency mentioned in OGN/OPS/CHEM/017 Cycle chemistry guidelines for start-up, Shutdown and layup practices" for more details. - All PM jobs to be done as planned - Reg. BCW p/p Refer "OGN/OPS/MECH/003 "O&M GUIDELINES FOR BCW PUMPS" NOTE: For Dry preserving boilers, preservation and hot draining sequence should be made part of shutdown checklist itself.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
BOILER SYSTEMS	AIR PRE-HEATER	•Ensure Operation of soot blower to clean the APH basket. •Ensure that air preheater soot blowers are not leaking water into ductwork afterwards. •Rotate air preheaters during the drying process or as prescribed by OEM. •Check for any seal rubbing.	•Perform visual inspection of air preheater including baskets, seals, rotors, and structural members.	•APH main motor and air motor along with support, Guide brg LOPs trial should be taken fortnightly •Oil sampling for oils for support and guide brg for moisture, MI every fortnightly preferably after every trial and PQ index / Wear Debris Analysis on quarterly basis.
BOILER SYSTEMS	ID FAN	•Ensure that the fan is in good working order and there are not any mechanical concerns about a return to service. •Fan cleaning should be coordinated to follow after any other duct or precipitator washing activities that could transport residue into the fan area.	•Sample, filter, or replace oil. •Perform duct inspection for any ash accumulation and same to be cleared.	•Fan trial should be taken every fortnightly along with LOPs and seal air fans.IGV operation also to be checked. •Oil sampling for moisture, NAS, MI fortnightly preferably after every trial.
BOILER SYSTEMS	SEAL AIR FANS AND OTHER DAMPER SEAL AIR FANS AND SCANNER AIR FANS	•Ensure that the fan is in good working order and there are not any mechanical concerns about a return to service. •Fan cleaning should be done for upkeep and healthiness of fans.	•Sample, filter, or replace oil	•All gate/damper seal air fans should be tried fortnightly.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
FUEL SYSTEMS AND AIR	MILLS	•For units which are mostly under RSD bunker levels should be maintained on lower side. •During shutdown of the unit, ensure that all pulverizes are completely emptied of coal	•Monitor CO readings at the silos for fire concerns •If the layup period extends to several days, consideration should be given to emptying the bunkers due to fire concerns or hardening of coal inside the bunker and causing rat hole afterwards.	•All mill LOPs trial should be taken fortnightly. •Mill lube oil sampling to be done for moisture and MI in minimum every 15 days and PQ index/ Wear Debris Analysis on quarterly basis. •Monthly mill trial should be taken. •Bunker temperature to be monitored for fire and recorded. •
FUEL SYSTEMS AND AIR	PA FANS	•Ensure that the fan is in good working order and there are not any mechanical concerns about a return to service. •Fan cleaning should be coordinated to follow after any other duct or precipitator washing activities that could transport residue into the fan area •Fan internal inspection to be done if vib> 4.5 mm/s	Perform duct inspection for ash accumulation.	•Fan trial should be taken every fortnightly along with LOPs and seal air fans. (or as prescribed by OEM). Blade pitch operation also to be checked. •Oil sampling for moisture, NAS, MI fortnightly preferably after every trial.
FUEL SYSTEMS AND AIR	FD FANS	•Ensure that the fan is in good working order and there are not any mechanical concerns about a return to service •Fan cleaning should be coordinated to follow after any other duct or precipitator washing activities that could transport residue into the fan area	Perform duct inspection for ash accumulation.	•Fan trial should be taken every fortnightly along with LOPs and seal air fans. Blade pitch operation also to be checked. •Oil sampling for moisture, NAS, MI fortnightly preferably after every trial.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
FUEL SYSTEMS AND AIR	HFO/LDO GUNS	•Guns should be kept in isolated condition with manual valves closed. •Any leaking oil gun inside the furnace should be attended.		•All oil guns should be scavenged fortnightly •Burner tilt operation to be checked fortnightly.
FUEL SYSTEMS AND AIR	ASH HANDLING (WET SYSTEM)	•Ensure that wet bottom ash hoppers are cleared of any clinkers or ash buildup in hoppers as soon as possible after shutdown of the unit •Inspect the wet ash removal systems to ensure hoppers, transport lines, etc. are emptied		All PM jobs to be done as planned
FUEL SYSTEMS AND AIR	ASH HANDLING (DRY SYSTEM)	•Ensure that all fly ash hoppers (precipitator, economizer hoppers, primary air, etc.) are completely empty of ash as soon as possible after shutdown of the unit.		
MILLING SYSTEM	RC FEEDERS	•Ensure that all feeder coal belts are emptied at shutdown with feeder inlet gates closed. •A thorough cleaning around the coal belts should be considered		•Empty Feeder trial to be taken every month with the help of C&I.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
STEAM TURBINE AND GENERATOR	TURBINE	• Once the turbine comes below 100 deg C turbine barring gear can be stopped. However it is advisable to keep TG on barring. • Extraction and drain valves should be opened and the lines and traps drained. The gland piping system should be drained and left in a dry condition. • Keep a steam seal on the turbine if hot standby status is required. If the condenser vacuum is not held, seal steam can be taken off the turbine • Apply dehumidified air to the turbine to ensure dry conditions are maintained. • OEM Manual for mandatory preservation needs to be adhered to.	• CF pumps trial to be taken on weekly basis along with auto changeovers.	• For dry preservation of turbine, RH should be measured every week for preservation. • If barring is stopped lube oil should be taken in service along with barring for 3hrs twice in the week. Along with all the pumps auto start checking. • Lube oil, control fluid, all OSU in turbine side like, LPBP, HPBP should be sampled every week for MI, Moisture and NAS. • Oil heaters should be kept in service and coolers may be isolated to maintain oil temperatures. • All purification of lube oil should be done as prescribed by OEM. • Lube oil system may be kept in service if required for seal oil system. • All purification and heating of control oil should be done as prescribed by OEM. • All CF system accumulator healthiness to be ensured on monthly basis and records to be kept.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
STEAM TURBINE AND GENERATOR	GENERATOR	• Generator casing pressure to be maintained as per normal running condition. • Seal oil system shall remain in service. • Ensure that bus cooling system dehumidification is properly functioning • If the layup period extends over a week, H2 coolers should be isolated.	• Monitor hydrogen purity and dew point to ensure that these are at acceptable level. • Ensure that generator liquid detectors are in service and operational. • Gas consumption should be recorded. • Seal oil temperatures should be maintained. • Primary water and gas pressure DP should be maintained. (H2 pressure should be more than pw pressure at generator inlet) • N2 capping pressure to be maintained on PW tank.	• Gas purity should be measured every day and dew point once in a week. • Seal oil sampling should be done for moisture, NAS and MI every week. • DC drive trials and schedule change over should be done as done during unit on-bar. • Bus duct pressurization system should be in service and offline dew point measurement should be done on weekly basis. • LLDs to be checked daily and recorded. • Records of all above activities to be maintained. • While taking trial of DMCW stagnant cooler water to be replaced with fresh water.
STEAM TURBINE AND GENERATOR	EXCITER	• If the layup period extends to more than a week, isolate the exciter coolers. • Exciter heater should be in auto and healthiness to be ensured.	All PM jobs to be done as planned	• While taking the trial of DMCW stagnant water inside the exciter cooler should be replaced by fresh one by lining up during the trial.
STEAM TURBINE AND GENERATOR	STATOR COOLING WATER	• Keep stator cooling water in service along with ensuring the stator cooling water is treated and is consistent with water chemistry guide lines.		• PW sampling should be done on weekly basis for conforming to chemistry guidelines • Schedule change over to be done.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
FEEDWATER AND CONDENSER	CONDENSER AND HOTWELL	• Please refer the OGN COS-ISO-00-OGN/OPS/CHEM/017 "Cycle chemistry guidelines for start-up, Shutdown and layup practices" for detailed procedure.	• During wet layup, ensure that good water chemistry is maintained • Please refer the OGN COS-ISO-00-OGN/OPS/CHEM/017 "Cycle chemistry guidelines for start-up, Shutdown and layup practices" for detailed procedure.	• Sampling to be done on weekly basis and records to be maintained. • Sampling protocol should be as per the OGN/OPS/CHEM/017 "Cycle chemistry guidelines for start-up, Shutdown and layup practices".
FEEDWATER AND CONDENSATE SYSTEM	FEEDWATER AND CONDENSATE SYSTEM	• Maintain the feed water and condensate systems in wet layup • If the layup period will extend for several days, establish a nitrogen blanket on the shell side of the feed water heaters. • If the layup period extends to several days, consideration should be given to going to a dry layup. • For timeline OGN/OPS/CHEM/017 may be referred.	• Maintain good water chemistry on the feed water and condensate systems during layup period • Maintain a slight positive pressure of nitrogen on feed water heaters during layup period • Please refer the OGN/OPS/CHEM/017 "Cycle chemistry guidelines for start-up, Shutdown and layup practices" for detailed procedure.	• Sampling to be done on weekly basis and records to be maintained.
FEEDWATER AND CONDENSATE	TDBFP	• Keep the boiler feed pump turbine on turning gear till casing temperature 85 deg C. • If the layup period extends to more than 7 days, barring can be stopped & LOPs to be made off. • Periodically rotate the turbine/pump rotor		• TDBFP MOT sampling to be done weekly for MI, moisture and NAS. • TDBFP to be taken on barring weekly for minimum 6 hours • All LOPs trial should be taken weekly. • Lube oil heater operation and centrifuging to be done regularly.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
FEEDWATER AND CONDENSATE	FEEDWATER HEATERS & DEAERATOR	• If shutdown is < 4 days, deaerator to be kept pressurized & temp to be maintained more than 110 deg c. • If shut down is extended to > 4 days then its vent to be closed & additional provision to be given for N2 purging & capping.	Water chemistry to be maintained as per normal unit running condition	• Water sample to be taken once in 15 days. • Water to be circulated by BFP once in 15 days.
BALANCE OF PLANT SYSTEMS	COOLING WATER SYSTEM	• Keep CW system in service till casing temp. Comes below 325 deg C. • For spring mounted TG deck barring to be stopped during draining or charging of CW system.	•	• CW sampling to be done weekly. • Each CW pump should be tried monthly such that water is circulated twice in a week for at least 2 hours. • During CW circulation chemical dosing to be done based on hold up volume.
BALANCE OF PLANT SYSTEMS	TGECW, SGECW AND ACW	• ACW system can be stopped 4-5 hours after unit shutdown. • SGECW pump can be stopped if water is not required for cooling purpose. • TGECW system to be kept in service as seal oil cooling is required.		• Each pump trial should be taken on weekly basis. • All valves operation should be checked every 15 days. • pH value should be monitored and maintained.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
BALANCE OF PLANT SYSTEMS	WATER SUPPLY AND WATER TREATMENT	• Keep Condensate Storage Tanks as it is.		• Sampling to be done every 15 days • CST p/p trial to be taken every month.
BALANCE OF PLANT SYSTEMS	IA/PA AIR SYSTEMS	• Boiler side IA/PA can be isolated soon as APH cools down to ambient temp. • Interconnection between stage-I & 2 to be kept open & IAC/ PAC can be stopped. • Ensure adequate measures are taken to reduce moisture in the air system. (Air dryer system, heat tracing, moisture drain systems in place etc.)		• Schedule changeovers to be done for compressor and driers. • MRC should also be tried every 15 days minimum.
BALANCE OF PLANT SYSTEMS	VACUUM PUMPS	• Liquid ring vacuum pumps should be drained while in preserved state.		• Trial should be taken on fortnightly basis by filling the liquid ring and draining after the trial. • Not more than 5 minutes trial should be taken in case of condenser vacuum pumps.(vacuum breaker to be kept open while trial)

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
BALANCE OF PLANT SYSTEMS	HFO/LDO PUMPS, RAW WATER P/PS ETC	•HFO LDO tanks to be kept as it is. •HFO tank temperatures to be maintained. •Firefighting equipments and systems to be kept in service.		•Fortnightly trial of HFO, LDO pumps should be taken. •Tank level should be monitored for any water ingress or leakages.
BALANCE OF PLANT SYSTEMS	FIRE PROTECTION SYSTEMS	•Ensure all Fire Protection Systems are in service •Ensure the Backup Fire Pump (Diesel, DC, etc.) is operational		•All diesel fire hydrant, sprinkler and booster pumps should be tried daily •Schedule change over to be done for all electrical fire water pumps.
BALANCE OF PLANT SYSTEMS	ACTUATORS	•If the layup period extends to more than 2 weeks, consideration should be given to operating critical plant actuators on a frequent basis		•All boiler, Turbine and Offsite electrical actuators should be tried at least every 15 days for remote operation.
ELECTRICAL SYSTEMS	SWITCHGEAR	•Switchgear should be kept warm and dry		• Healthiness of auto changeover of incomers and bus couplers should be tried every month along with records.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
ELECTRICAL SYSTEMS	TRANSFORMERS	- Transformers to be kept charged. - The heater strips located in the control cabinets are to remain energized in order to prevent damage to control equipment by moisture condensation. - Auto changeover of cooling system of GT/UT	- After shutdown of the unit, verify proper operation of the control cabinet strip heaters - After shutdown of the unit, verify there are no major oil leaks. - For GT/UT auto changeover of cooler supply to be checked.	- The transformers are to remain oil filled with the proper oil levels - Transformer oils to be sampled as per electrical OGN/OPS/ELEC/020 - GT and UT cooling system trial to be taken with EMD on fortnightly along with auto changeover of supplies. - Lightning arrester to be checked for healthiness with EMD.
ELECTRICAL SYSTEMS	MOTORS WITH INTERNAL HEATERS	- Ensure motor heaters are in service - Ensure all motors are lubricated or oil reservoirs have sufficient oil during shutdown - If the layup period will extend to more than two weeks, consideration should be given to rotating large motors		- IR value measurement to be done before every trial. - Motor space heaters healthiness to be ensured along with maintenance. - In case of MDBFP, motor trial should be taken on monthly basis if boiler, deaerator, feed water is preserved in wet preservation state.
ELECTRICAL SYSTEMS	MOTORS WITHOUT INTERNAL HEATERS	Should be kept As it is		Trial should be taken on fortnightly basis
ELECTRICAL SYSTEMS	DG SETS AND DC DRIVES	To be kept As it is.		As long as turbine is on barring or hydrogen is in the generator casing, DG and DC drives trial should be taken on daily basis.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
ELECTRICAL SYSTEMS	BATTERIES AND UPS	• Maintain batteries on float charge. • Keep UPS System in service	• If the layup extends to several days, consider measuring battery and charger output	• Daily monitoring should be done and recorded in log book. • PM should be done as per planning.
ELECTRICAL SYSTEMS	DCS AND CONTROL SYSTEMS	• Keep the DCS system in service		CER temp & humidity to be monitored
ENVIRONMENTAL CONTROLS	ESP	• Keep fields in off condition along with BABCON and RAPCON panel. • Hopper emptying to be ensured after shutdown. • Consideration may be given to use rappers (higher intensity on rappers) on shutdown to clean precipitator plates and wires • If the precipitator Internals have a high ash buildup, considerations should be given to washing the precipitator internals.		• Rapping motors trial to be done at least once fortnightly for min. 30 minutes. • Shaft insulator and support insulator heating mechanism should be always on. Note: Shaft insulator heating may be stopped if it can be started before 48 hours of unit light-up (light-up information in known before 48 hrs). Trial should be taken on monthly basis. Hopper heaters are to be checked one in a month.

7.5 EQUIPMENT LAY-UP MATRIX				
SYSTEM	SUB SYSTEM	ACTION FOR LAYUP	INSPECTION AND MONITORING ACTIVITIES	SAMPLING AND UPKEEP
CONDENSATE SYSTEM	CPU SERVICE VESSEL	• Air mixing and rinsing of CPU service vessels to be done on fortnightly basis. • Associated systems like regeneration systems, spare resins should be checked periodically for healthiness.		
ASH HANDLING SYSTEM	BOTTOM ASH HOPPER/ ECONOMISER HOPPER/APH HOPPER	• Condition of refractory and mother plate (S panel sealing needed along with Scaffolding erection in bottom ash hopper by BMD) • Refractory cooling header and BAH flushingheader/ quenching nozzle condition • Seal trough condition • Jet pump nozzle condition • Any equipment needs urgent maintenance • Eco/APH hopper emptiness	• Check the condition • Check condition of the headers and nozzles • Check for any choking of drains • Check for any choking/enlargement • Check the repair required Check for ash accumulation/entrapped foreign material	• Repair to the extent the duration permits. • Cleaning/replacement • Cleaning of seal trough and removal of choking/jamming of drain valves • Replacement if required. • Attend the defect. Clear the choking and ensure emptiness
ASH HANDLING SYSTEM	FLY ASH EVACUATION	• Emptiness of ESP hoppers • Check for air ingress points of ESP hoppers • Vacuum headers, DAES valves' and associated pipelines' condition • Condition of the equipments e.g. vacuum pumps, wetting head, bag filters, pulsejet system, fluidising system etc. • TAC, IAC and other rotating equipments.	• Check for ash accumulation/entrapped foreign material • Locate the leaking ESP manhole doors • Check for any defect in the Vacuum headers, DAES valves and associated pipelines • Check for any defect (including the already identified ones) in the equipments for FA evacuation system. • Check for its freeness	• Ensure emptiness and prepare signed protocol • Change ropes of the manhole doors and ensure the air ingress is arrested • Repair / replacement of defective Vacuum headers and DAES valves and associated pipelines. • Attend the defect. • Trial run of the rotating equipments once in 15 days
ASH HANDLING SYSTEM	ASH SLURRY DISPOSAL	• Condition of the pipelines • Pumps' condition	• Check for any leakage/erosion • Check for free rotation	• Rotation/repair/replacement of pipes and bends. • Greasing / manual rotation to be done

Guidelines for unit reserve shutdown

ASH HANDLING SYSTEM	FA SILO	Condition of the equipments e.g. vent fans, bagfilters, pulsejet system, fluidising system, HMDC, Rotary feeders, Condition of target box and pipelines	- Check for any defect - Check for any defect	- Attend the defect. - Attend the defect.
HOUSE KEEPING	HOUSE KEEPING	All the areas including boiler, turbine and offsite should be well maintained and cleaned; any presence of unsafe and flammable material including grass should be removed on priority.		
COAL HANDLING PLANT	UNLOADING AREA (WAGON TIPPLER & WAGON)	In case for no rake receipt, no load operation for these equipment once a day to be carried out for half an hour	During No Load running, inspection and monitoring for abnormal sound, leakages, drive current etc to be done.	Oil sample for Hydraulic Oil system may be taken if oil purifier not installed in circuit.
COAL HANDLING PLANT	YARD ZONE	In addition to above, proper compaction of coal in yard to be ensured and water spray for preventing spontaneous combustion.	Check periodically for any smoldering in coal piles.	Various Yard mapping activities including age analysis, CO Monitoring, Compaction Index as per OGN to be followed.
COAL HANDLING PLANT	CONVEYOR SYSTEM & FLAP GATES	For post crushing path, no load operation of half an hour of each conveyor on daily basis to be ensured. Flap gates changeover to be checked.	During No Load running, inspection and monitoring for abnormal sound, leakages, drive current, ease of flap gate changeover etc to be done.	NA

Primary responsibility of preservation of units & equipment lies with SCE/UCE. It is responsibility of UCE to give samples of oil /water at mentioned intervals & take trial run of equipment as per the prescribed time interval. Concerned maintenance department will facilitate operation department for preservation activities.

8.0	RECORDS OF TEST & VARIOUS SAMPLES
	All the records with respect to chemical samples of oil and water, preservation status and schedule trials to be maintained and a monthly “Preservation and upkeep Report” should be generated and exceptions to be discussed in DPM. Any exception in preservation and testing of operability of various equipment discussed in layup matrix should be attended on priority.
9.0	RESPONSIBILITY
	HOD (O) is responsible for implementation of this LMI
10.0	REFERENCE DOCUMENT
	This LMI is based on operation Guidance Notes for guidelines on units under reserve shutdown OGN/OPS/SYST/038 Rev No.1, April 2023
11.0	REVIEW
	The HOD will be responsible for reviewing this LMI after 2 years or as necessary.

EQUIPMENTS TRAIL RUN AND SAMPLING SCHEDULE DURING UNIT RESERVE SHUTDOWN

EQUIPMENT STOPPAGE SEQUENCE AFTER UNIT STOPPAGE

Unit:

Date:

Unit stoppage Time:

UNIT AUXILIARY TO BE STOPPED	Time Since Unit Outage			REMARKS/CRITERIA
	Target	Actual		
		Time	Duration	
One Set of ID/FD fan	15 min			-
One BCW p/p	15 min			-
One ACW p/p	15 min			To be Stopped based on ambient temp.& Auxiliaries temp
One TGECW p/p	15 min			To be Stopped based on ambient temp.& Auxiliaries temp
One CEP	15 min			Hotwell level to be monitored
CT fan	30 min			All fans/Alternate fans to be stopped as per ambient temp.
DM make up pump	1 hr			To be Run based on DM make up requirement of other unit
MDBFP	3 hrs(Approx.)			Intermittent start req. to maintain drum level
Vacuum P/P	2 hrs(Approx.)			Vacuum killing after Main TG barring stable
2nd CEP	3 hrs			After seal steam isolation. Observe exhaust hood temperature. CEP to be started as per water requirement in Deaerator.
One IAC/PAC	4 hrs			To be stopped based on other Unit req./Observe I.A. pr.
2 nd ID fan	5 hrs(Approx.)			When flue gas temperature before APH <205°C
2 nd FD fan	5 hrs(Approx.)			When flue gas temperature before APH <205°C
All ESP fields	5 hrs(Approx)			After To be Stopped of 2 nd set ID/FD fans. Watch opacity, if more due to natural draught keep 1 st /2 nd fields i/s
Boiler Side LOP	5 hrs			After 1hr of Fan/Mill To be Stopped.
2 nd ACW p/p	8 hrs			May be stopped earlier based on ambient temp. & auxiliaries temp
2 nd BCW p/p	14 hrs (Approx.)			When suction manifold temp <93°C (or as per Dry preservation SOP)
AC scanner air fan	24 hrs			When furnace temperature <149 °C. Ensure DC scanner fan supply off
SGECW p/p	30 hrs(approx.)			When CC pump suction manifold temp <65°C .Before To be Stopped ensure SD-21&22 are tagged out
ESP Rapping/heaters	24 hrs			After ensuring complete evacuation of hoppers
1 st &2 nd CW p/p	48 hrs/HP casing 325°C (whichever is later)			-

RSD TRIALS CHECKLIST

TRIAL SCHEDULE (DAILY)	DURATION	COMPLIANCE
DC SCANNER FAN	10MIN	
APH AIR MOTOR	10MIN	
MAIN TG DCEOP	10MIN	
TDBFP-A DC EOP	10MIN	
TDBFP-B DC EOP	10MIN	
MDBFP AOP	1HOURS	
DC SOP	2 MIN	
DIESEL FIRE HYDRANT PUMP	10MIN	
DIESEL FIRE BOOSTER PUMP	10MIN	
DIESEL FIRE SPRAY PUMP	10MIN	
MAIN DG SET	15MIN	
MOT CENTRIFUGE	12HRS	
TDBFP CENTRIFUGE	12HRS	
TRIAL SCHEDULE (WEEKLY)	DURATION	
ALL EQUIPMENTS AS PER SCHEDULE C/O REGISTER	-	
TDBFP BARRING	SIX HOURS	
MAIN TG BARRING TWICE IN A WEEK	3 HOURS	
CONTROL FLUID PUMP	1HOUR	
AHP IAC	30MIN	
AHP TAC	30MIN	
AHP VACUUM P/P	30MIN	
ESP FLUID.BLOWER	30MIN	
BASP	30MIN	
HCSO	30MIN	
AHP AREA DRAIN P/P'S	30MIN	
BAHP	30MIN	
BALP	30MIN	
FAHP	30MIN	
ECO. P/P	30MIN	
FLUSHING P/P	30MIN	
BASP SEAL WATER P/P	30MIN	
CLINKER GRINDERS	30MIN	
AWRS P/P	30MIN	
PIPRI P/P & ITS LUB WATER P/P	30MIN	
FA SILO COND.P/P	30MIN	
FA SILO FLUID.BLWR	30MIN	
TRIAL SCHEDULE (15DAYS)	DURATION	
ID FAN	15MIN	
FD FAN	15MIN	
PA FAN	15MIN	
SEAL AIR FAN	15MIN	
ALL MILLS	10MIN	
ALL FEEDERS	10MIN	

AC SCANNER FAN	ALONG WITH ID/FD/PA TRIAL	
APH MAIN MOTORS	30MIN	
CC PUMP/BRP	30MIN	
BURNER TILT OPERATION	FULL TRAVEL	
ESP RAPPING SYSTEM	30MIN	
MDBFP SCOOP&RAPID STARTUP SOLENOID OPERATION	FULL TRAVEL	
MDBFP	15MIN	
PW PUMP CHANGE OVER	-	
CW PUMP	AS PER CHEMISTRY REQ.	
ACW PUMP	AS PER CHEMISTRY REQ.	
TRIAL SCHEDULE (MONTH)	DURATION	
MAIN TURBINE SV/CV	FULL TRAVEL	
HP/LPBP V/V'S	FULL TRAVEL	
TDBFP SV/CV/ACV	FULL TRAVEL	
CF SYSTEM ACCUMULATOR CHECKING (ALONG WITH TMD)	-	
TGECW CIRCULATION IN GENERATOR & EXCITER COOLERS WITH VENTING	-	
MILLS	10MIN	
RC FEEDERS	10MIN	
CT FANS	15MIN	
ACTUATORS TRIAL (AS PER CRITICAL ACTUATORS LIST)	-	
AUTO C/O OF I/C AND B/C IN SWGRS (AS PER CRITICAL SWGR LIST)	-	
GT/UT COOLING SYSTEM (ALONG WITH EMD)	30MIN	
LIGHTENING ARRESTOR CHECKING (ALONG WITH EMD)	-	
ESP HOPPER HEATERS	1HOUR	
ESP SHAFT/SUPPORT INSULATOR HEATER	1HOUR	